Stanley Nelavala

Backend Software Engineer
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Huntington, IN | Open to relocate anywhere

Summary

Backend engineer who owns systems end to end. At a 10-person hospital operations startup I designed the service-to-service API that every platform module uses, cut a deep catalog query from 285 ms to 0.20 ms with keyset paging, and fixed a regular expression denial-of-service bug. Public Java, Rust, C# and Python services with idempotency, outboxes and failure labs. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Fixed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s; the new linear pattern matched the old results on 60,000 random inputs.
- Cut the planning Items page from 3 API calls and 162 KB to 2 calls and 37 KB with typeahead search and cursor paging.
- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine that places rooms on a 0.5 ft grid with doors, exits and flow lines; 220+ offline tests on the API.
- Cut dev and acceptance server and database (ECS, RDS) runtime to about 53 hours a week instead of 168 with automated weekday start and stop. Standardized malware scanning of file uploads (GuardDuty) across 9 app modules.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session. Led labs and office hours for Python, NLP and Parallel Programming.

Projects (public code, tests)

Payments Ledger Service | *Java 25, Spring Boot, PostgreSQL, Flyway, Kafka API, Prometheus, Grafana, k6* GitHub

- Double-entry ledger (authorize, capture, void, refund, transfer) with idempotency keys and a transactional outbox to a Kafka API (Redpanda); property tests and a k6 check that no money is created or lost.
- Three user-facing SLIs with 28-day error budgets and six multi-window burn-rate alerts tested with promtool; a scripted failure lab (Kafka down, Postgres restart, relay crash, poison event, latency injection) with balance checks after each.

Webhook and Connector Gateway | *Rust, axum, tokio, PostgreSQL, SQLite, sqlx, OpenTelemetry, k6* GitHub

- Rust gateway: HMAC-SHA256 webhooks verified in constant time, per-tenant rate limits and idempotency, durable store before delivery, retries with backoff, dead letters with replay and circuit breakers.
- Postgres store (FOR UPDATE SKIP LOCKED claim) beside SQLite behind one store port; the same 88 tests pass on both backends.

Card Dispute Service | *C#, .NET 10, ASP.NET Core, EF Core, PostgreSQL, xUnit, CloudFormation* GitHub

- Dispute state machine where concurrent decisions on one case produce exactly one winner (PostgreSQL xmin optimistic concurrency); idempotent create and keyset paging tested with ties, inserts and deletes; tests written first (30 unit, 32 integration).

Skills

Languages: Python, Java, Rust, Go, C#, SQL, TypeScript
Backend: FastAPI, Spring Boot, axum and tokio, ASP.NET Core, EF Core, REST, gRPC, idempotency, transactional outbox
Data and messaging: PostgreSQL, Redis, Kafka API (Redpanda), SQLite, migrations (Alembic, Flyway, sqlx), pgvector
Cloud and reliability: AWS (ECS, RDS, S3, SQS), Docker, GitHub Actions, OpenTelemetry, Prometheus, Grafana, SLOs
Testing: pytest, JUnit 5, Testcontainers, xUnit, property tests, k6, Playwright

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)